BERKE DURU

Computer Engineer • Software Development • AI & Computer Vision

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ABOUT ME

I'm a Computer Engineering graduate from Yaşar University, currently pursuing a master's degree in Computer Engineering. I have hands-on experience across software development, artificial intelligence, computer vision, and embedded systems. I have worked on desktop and cross-platform applications, developer tools, autonomous systems, and integrated software solutions involving networking and workflow automation. My projects include real-time computer vision and autonomous UAV systems running on NVIDIA Jetson platforms, as well as an integrated order, customer communication, and workflow management system connecting multiple user roles and external services. I have also developed open-source tools for computer vision workflows and Unity-based game and VR applications. I'm interested in roles where I can apply my experience across application development, intelligent systems, and real-world problem solving.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, C#, JavaScript

Software & Application Development: Application Development, System Integration, Workflow Automation

AI & Machine Learning: Machine Learning, Deep Learning, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

Computer Vision & Edge AI: OpenCV, Ultralytics YOLO, Object Detection & Tracking, TensorRT, ONNX, CUDA, DeepStream, GStreamer, Model Training & Fine-tuning

Embedded & Autonomous Systems: NVIDIA Jetson, JetPack SDK, Raspberry Pi, ROS, ArduPilot, MAVLink, Gazebo

Game & XR Development: Unity, OpenXR, XR Interaction Toolkit

Systems & Networking: Linux (Ubuntu), Docker, TCP/IP, UDP, RTSP, Git/GitHub, SQL

Cloud & Platforms: Firebase, Cloudflare

Additional Tools: Google Colab, PyQt, QGIS/PyQGIS, Selenium

EXPERIENCE

İzmir Innovation and Technology Inc. — Directorate of AI & Smart City Technologies

R&D Engineering Intern | Jul 2025 - Aug 2025 | GitHub Project

- Developed an end-to-end computer vision workflow for detecting potential new or unauthorized construction by comparing multi-date geospatial imagery.
- Automated geospatial data collection using Python and Selenium, and processed the resulting data with QGIS and PyQGIS.
- Applied difference masking and morphological operations to identify structural changes and generated coordinate-based outputs for further analysis.

EDUCATION

Yaşar University — İzmir, Turkey

Master's Degree in Computer Engineering (Thesis) | 2026 - Present | 100% Scholarship

Yaşar University — İzmir, Turkey

Bachelor's Degree in Computer Engineering | 2022 - 2026 | GPA: 3.31 / 4.00 | 75% Scholarship

LANGUAGES

English: B2

PROJECTS & RESEARCH

Integrated Printing & Document Services Operations Platform — September 2026

JavaScript, Firebase, Cloudflare, System Integration, Workflow Automation

- Developed an integrated operations platform for a **printing and document services business**, covering the workflow from customer file and order intake to document preparation, pricing, payment tracking, production, and print dispatch.
- Connected in-store workflows with web/mobile, WhatsApp, and e-mail order channels, allowing customer requests and files to enter the same centralized order process.
- Implemented centralized product/pricing, customer, order, and payment management together with synchronized multi-workstation workflows for customer service, document preparation, cashier, and administration.

TEKNOFEST Fighter UAV — Computer Vision & Autonomous Systems — May 2025 – Aug 2026

Anafarta UAV Team | Python, YOLO, NVIDIA Jetson, TensorRT, DeepStream, GStreamer, ArduPilot, MAVLink

- Developed the end-to-end aircraft detection workflow, including dataset preparation, annotation, YOLO model training, evaluation, inference, and Jetson deployment.
- Built real-time DeepStream/GStreamer pipelines on NVIDIA Jetson Orin platforms for camera/UDP input, TensorRT inference, video streaming, and recording.
- Worked on integrating computer vision outputs with UAV autonomy and telemetry workflows using ArduPilot and MAVLink.

Path Planning & Dynamic Obstacle Avoidance for Autonomous Systems — 2026

University-Funded Research Project (BAP)

- Worked on path planning and dynamic obstacle avoidance methods for autonomous systems.
- Developed and evaluated APF-based navigation approaches for motion planning in environments containing obstacles.

Olive Yield Estimation & Digital Mapping — 2025

Research Project Proposal (BAP) | Team Project

- Prepared a research project proposal for olive tree detection and yield estimation using high-resolution UAV aerial imagery and YOLO-based computer vision methods.
- Designed a proposed digital agriculture workflow combining tree detection, location mapping, tree counts, and yield estimation.

Game & XR Development

C#, Unity, OpenXR, XR Interaction Toolkit, Meta XR

- Developed multiple Unity-based game projects involving gameplay mechanics, UI systems, animations, player interactions, and game logic.
- Developed an interactive VR project using OpenXR and XR Interaction Toolkit with controller interaction, locomotion, object manipulation, weapon/reload systems, and AI navigation.
- Contributed to additional Unity projects including a card-opening game with interactive pack-opening and animation systems.

HONORS & AWARDS

- **GBYF 2026 — Top 6 Nationwide Finalist:** Finalist in the Academic and Information Systems Applications category at the Genç Beyinler Yeni Fikirler Project Competition.
- **EFES Defense Science & Technology Exhibition 2026 — Top 10:** Selected among the Top 10 University Defense Projects with the autonomous UAV project — Anafarta UAV Team.
- **TEKNOFEST 2026 Fighter UAV Competition — Finalist:** Anafarta UAV Team.
- **TEKNOFEST 2025 Fighter UAV Competition — Finalist:** Anafarta UAV Team.